

Competitive Dynamics of Trading Fee Structures in Limit Order Markets

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Abstract

Trading venues use fees to compete for order flow. Currently, two fee regimes are used by most lit exchanges: the make-take and the inverted make-take-fees¹. This study develops a theoretical model to examine the dynamics of both fee regimes within a competitive framework. We show that inverted venues experience better market quality and more trading activity at the inner quotes, whereas make-take venues experience the same at the outer quotes.

Key words: Market microstructure, make take-fees, inverted make take-fees, order submission probability, order choice, intermarket competition, market equilibrium, limit order book, Markov chains

¹Also referred to as the take-make-fees

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1 Introduction

Currently, two dominant fee regimes prevail: the widely adopted make-take-fees and the less common take-make-fees. In the make-take model, exchanges charge fees to liquidity takers and offer rebates to liquidity providers. Conversely, in the take-make model, exchanges pay rebates to liquidity takers and charge fees to liquidity providers. Exchanges employ both fee regimes to compete for trading volume.

The emergence of trading fees were most likely a reaction to retain a competitive advantage in an increasingly fragmented market. Historically, exchanges made their profit partly by charging access fees to all traders. As more trading venues entered the space, trading fees were driven down to compete for liquidity ([Aitken, Chen, and Foley \(2017\)](#)). The emergence of dark pools² at the start of the 1980s further fueled the competition between trading venues, as the market share of dark pools steadily remained at 3 - 5% for the next 20 years ([Zhu \(2012\)](#)). In response to this fragmentation, make-fees were introduced to attract liquidity. The Island ECN was the first to implement a make-fee in 1997, which proved to be an immense success, as their market share of reported NASDAQ trades increased from 3% in 1997 to 13% in 1999 ([Division of Trading and Markets, SEC \(2014\)](#)). This success caused other exchanges to adopt similar fee regimes by passing a portion of their access fees as rebates to liquidity providers. By 2003, most (US) ECN's paid a rebate for non marketable limit orders ([Federal register SEC \(2004\)](#)). In 2007 the SEC passed Regulation NMS (National Market System), which allowed investors to bypass public exchanges to gain price improvements, which shot up dark trading to 10% by 2010. The increased relevance of dark pools, combined with a saturation of make-take-fee regimes, likely contributed to the emergence of take-make exchanges. Nasdaq was the first to introduce take-make-fees on the BX exchange in 2009. Since then, other major US platforms have offered this alternative pricing structure, including EDGA and BATS BYX (now part of CBOE).

[Ito, Firmin, and Comerford \(2017\)](#) demonstrated that the volume at take-make venues consistently rise throughout the day with a sharp rise near close. [Comerton-Forde,](#)

²dark pools are mentioned to provide context for the introduction of maker-taker fees; this topic is not explored further

Grégoire, and Zhong (2019) observed that inverted trading was paired with increased liquidity provision, which they attributed to a substitution effect with dark pools due to sub tick trading.³ Furthermore, Battalio, Corwin, and Jennings (2013) show that orders routed to inverted venues gain queue priority. Additionally, empirical evidence by Skjeltorp, Sojli, and Tham (2011) points out that algorithmic trading volume is higher on the inverted Nasdaq BX exchange compared to the conventional Nasdaq exchange, suggesting that the coexistence of various make/take-fee schedules may serve to screen different types of investors.

However, Angel, Harris, and Spatt (2010) noted that make take pricing can "distort order routing, aggravate agency pay-offs [...], produce fraudulent trades, and quotes that do not represent trading costs". Battalio et al. (2013) further emphasize the conflict of interest between brokers and their clients, as brokers are incentivized to route non marketable limit orders to venues with the highest rebates, regardless of execution quality.

Despite the extensive research body on trading fees and inter-market competition, limited theoretical research has been conducted on the dynamics between make-take and take-make venues at different price levels. This paper aims to frame these using a similar approach by Panayides, Rindi, and Ingrid (2023). The approach is grounded in a competition model between two identical limit order books: first, both order books are examined without fees to establish a benchmark understanding of market dynamics. Subsequently, both order books are examined with different fee regimes. The latest equity fee schedules from major U.S. exchanges are used as a reference. Concretely, the standard (non-tiered) maker taker fees from NYSE, NASDAQ and IEX are utilized and the fee schedules of each of these exchanges are compared to the fee schedules of the inverted maker taker fees of EDGA, BATS Y and NASDAQ BX.

This study shows that inverted venues experience better market quality and more trading activity at the inner quotes, whereas make-take venues experience the same at the outer quotes. This is consistent with Battalio et al. (2013), who found empirical evidence that Nasdaq BX and EDGA exhibit nearly three times as much displayed limit

³This was mainly due to the implementation of the trade-at rule, requiring all dark markets to provide a meaningful price improvement when executing dark orders

orders at the inner quotes. Furthermore, inverted trading is paired with increased overall trading activity, which compliments empirical observations by [Comerton-Forde et al. \(2019\)](#). This study also aligns with [Chao, Yao, and Ye \(2018\)](#), indicating that - all else being equal - liquidity providers have a considerably higher preference for take-make venues compared to make-take venues. Since traders can always undercut one another by placing limit orders at the inner quotes of the take-make venue. This is contrary to expectations since most trading activity typically persists at the make-take venues. Though this flattens out (i.e. migration from take-make to make-take) when make-fees widen and take-fees become narrower between exchanges.

Remark that in reality, trading fees are a complex matter which often depend on the added volume⁴, the type of order⁵ and the type of security. For equities, the fee could also depend on which network the security is quoted.⁶ The complexity further increases when intermediaries are taken into account⁷. Additionally, the lack of a defined 'end date' for most equity instruments introduces additional discrepancies. Therefore, derivatives, which often have explicit expiration dates, could provide a more appropriate basis for comparison. The extent of the limitations mentioned above could partially explain the disparity between this model's prediction and real-world outcomes, which poses interesting factors to consider for further research.

⁴The fees are usually 'tiered'. A higher tier entails adding more liquidity/removing volume and results in more favourable fees or rebates. Two metrics are often used for tiered based pricing : (1) added daily volume (ADV), a metric for the number of added daily shares and (2) adding/removing the percentage of total consolidated volume (TCV).

⁵For example, an exchange can give rebates for displayed liquidity while at the same time charging fees for non displayed liquidity (i.e. hidden orders). Other order types that might have differing fees include: Mid-Point liquidity (MPL) orders, orders at close/open, outside trading hours (OTH) etc

⁶In the US, real time market data are distributed on three networks: Tape A, B and C. Trades and quotes listed on NYSE are on Tape A, listings on Nasdaq and OTC securities on Tape C, and the rest of the exchanges are on Tape B. Tape A and B are known as the consolidated Tape and use the same Security Information processors (SIPs) for US equities.

⁷As noted earlier, most trades are posted at brokers. Who then subsequently route the orders to exchanges or market makers. A Conflict of interest might arise between brokers and their clients due to rebates that exchanges have to offer, or payment for order flow that market makers offer to brokers.

2 Literature review

Make and take-fees are controversial. Extensive research using theoretical models and descriptive statistics have been conducted to shed light on how various fee regimes influence the decision making of market participants and market quality. On the one hand, there are studies which point out that make and take-fees can improve pricing efficiency and is are paired with better market quality ([Malinova and Park \(2011\)](#), [Battalio et al. \(2013\)](#), [Comerton-Forde et al. \(2019\)](#)). On the other hand, it has been suggested that such fees aggravate agency problems and add to complexity [Angel et al. \(2010\)](#), [Colliard and Foucault \(2011\)](#), [Battalio et al. \(2013\)](#). The controversy has led the SEC to conduct a study on their own. In 2014 a pilot study was announced under Rule 610 to study the effects of the make-take-fees on market quality. However in 2020, this study was appealed by the court arguing the pilot was beyond the SEC's authority. Following the appeal, subsequent research has been done to frame the pilot either by making predictions using sample statistics or theoretical models. [Comerton-Forde et al. \(2019\)](#) frame this pilot study by utilizing data from the Tick Size Pilot, which ran from 2016 - 2018. They observed that increased activity on inverted venues led to enhanced market quality and improved liquidity provision, attributed to sub-tick price improvements. These findings support the proposition that inverted trading facilitates greater liquidity provision at the inner quotes through narrower cum-fee price grids. [Panayides et al. \(2023\)](#) developed a theoretical model and employed descriptive statistics to compare the effects of make-take-fee changes on BATS and Turquoise. They found that lowering trading fees is paired with improved market quality in make-take venues. This study shows that the same is true for take-make venues

This paper has its roots to [Parlour \(1998\)](#), who described a model implementing backward induction to find the order submission equilibrium in a finite period trading game. This model lays an important foundation for further theoretical research. Subsequent studies have built upon this model by incorporating cancellable orders and exogenous price movements [Goettler, Parlour, and Rajan \(2005\)](#), trading fees [Colliard and Foucault \(2011\)](#) and intermarket competition [Panayides et al. \(2023\)](#), the latter of which this study relates to most closely.

3 Set up

The trading game is structured into $T = 3$ periods. In each period $t_z = t_1, \dots, t_{T=3}$ a single risk neutral trader arrives, characterized by their gamma type (γ), which is i.i.d. drawn from the uniform distribution (γ) $\sim [0, 2]$ that is centered around the fundamental value $AV = 1$ of the traded asset. The distribution represents the extend to which a trader is willing to trade, where a γ -value of 0 indicates a low willingness to trade and a γ -value of 2 indicates a high willingness to trade. Hence, γ can be seen as a proxy for the trader's personal asset value.

There are two identical venues labeled Exchange I and Exchange II that are indexed by $j = I, II$. Both exchanges operate on a price grid with n price levels on both bid and ask side. The bid and ask prices are centered around the fundamental asset value AV with price increments of $\mu = 0.01$ such that each bid and ask price at each price level $i = 1, 2 \dots n$ equals $B_i^j = AV + \mu \cdot (\frac{1}{2} - i)$ and $A_i^j = AV - \mu \cdot (\frac{1}{2} - i)$ respectively. The order book of both exchanges has $n = 2$ price levels on each side such that the price grid P^j is defined as the concatenation of a reversed vector of ask prices A_i^j and a vector of bid prices B_i^j :

$$P^j = A^{j, \text{rev}} \oplus B^j = (A_2^j, A_1^j, B_1^j, B_2^j) = (1.015, 1.005, 0.995, 0.985)$$

The state of the order books of exchange j at time t_z is denoted by the vector $OB_{t_z}^j = \{lo_{t_z}(P_i^j)\}$, where $lo_{t_z}(P_i^j)$ is the depth of the order book at price P_i^j . The arriving trader at each period t_z observes the current state of the order books and chooses a strategy with a non zero execution probability among a set of strategies $ST_{t_z} = \{LO_{t_z}(P_i^j), MO_{t_z}(P_i^j), NT_{t_z}\}$ which consists of a limit order buy $LO_{t_z}(B_i^j)$, a limit order sell $LO_{t_z}(A_i^j)$, market order buy $MO_{t_z}(A_i^j)$, a market order sell $MO_{t_z}(B_i^j)$ or not trade NT_{t_z} . Limit orders are executed on a time-price priority basis within exchanges and cum fee price priority across exchanges⁸. The execution probability is equally distributed among limit orders that are placed at the same cum fee price across exchanges. Once placed, limit orders cannot be cancelled.

⁸The cum fee market order price to be precise

An exchange can either choose for a make-take-fee regime, a take-make-fee regime or implement no fees at all. Under make-take-fees, limit order traders are rewarded a positive make-fee ($MF^j > 0$) when their orders are executed, whereas market order traders incur a positive take-fee ($TF^j > 0$). In contrast, under take-make-fees, limit order traders pay a positive make-fee ($MF^j < 0$), whereas market order traders receive a negative take-fee ($TF^j < 0$). An exchange will structure their fees such that $(TF^j - MF^j) \geq 0$, simply because they need to be profitable in order to operate. Furthermore, make and take-fees are assumed to be restricted to 0.003 based on Rule 610.

4 Computing the Order Submission Equilibrium

The computation of the order submission equilibrium consists of two parts. In the first part, all the possible order book states are mapped considering the current state of the order books. In the second part, the probability of each state is determined by comparing payoffs from each order at t_3 and methodically working backwards to t_1 .

4.1 Order book states with a non zero execution probability

At t_1 , traders will only be able to submit limit orders. At t_2 , traders can either submit limit orders or market orders. In the last period t_3 , traders can only submit market orders. In all periods, traders can decide to not trade. In each period, there is a possibility that a particular order is submitted if it has a non-zero execution probability. For limit orders, this entails some possibility the order will be executed against a market order in the subsequent periods. For market orders, a non-zero execution probability implies the existence of a corresponding limit order that was placed in the previous periods. NT is always a possibility, regardless of the opening market and the current period. The logic behind the non-zero execution probability for limit and market orders is described more detail in the following subsections.

4.1.1 Execution probability for limit orders

A limit order has a non-zero execution probability if there is a possibility that a market order is placed against that limit order in one of the subsequent periods. Whether this will be the case depends on the state of the opening books, the cum-fee market order prices, and the number of subsequent periods. Concretely, a limit order is placed at a certain price P_i^{j*} in t_z if the number of limit orders at a better cum-fee market order price across all exchanges plus the number of limit orders at an equal price cum-take-fees within the exchange j is strictly less than the time outstanding $T - t_z$.

Execution probability for *LOB*⁹

$$\Pr_{exec}(LO_{t_z}(B_i^{j*})) > 0 \iff \sum_{i=1}^n \sum_{j=1}^n lo_{t_{z-1}}(B_i^j : B_i^j - TF^j < B_i^{j*} - TF^{j*}) + lo_{t_{z-1}}(B_i^{j*}) < T - t_z$$

Execution probability for *LOS*

$$\Pr_{exec}(LO_{t_z}(A_i^{j*})) > 0 \iff \sum_{i=1}^n \sum_{j=1}^n lo_{t_{z-1}}(A_i^j : A_i^j + TF^j < A_i^{j*} + TF^{j*}) + lo_{t_{z-1}}(A_i^{j*}) < T - t_z$$

4.1.2 Execution probability for market orders

A market order will always be executed if at least one limit order at the same side exists, otherwise it will never be executed.¹⁰ If multiple limit orders exist across both markets, then the market order is placed at the exchange with the best cum fee price for which a corresponding limit order exists. When the best available cum fee price is the same at both exchanges, then the trader can choose to place his market order at either one. A *MOB* is placed at exchange j in time t at the lowest cum fee ask price A_i^{j*} for which a limit order exists. There is an absolute certainty that the market order gets executed at that price. The execution probability for a *MOS* follows the same reasoning

Execution probability for *MOB*

⁹e.g. if opening books are $[[0, 0, 3, 0], [0, 0, 1, 1]]$ with 3 periods remaining and no fees. Then a *LOB* can be placed at the inner price of exchange II

¹⁰The execution probability of market orders follows a degenerative distribution, i.e. the probability is either 0 (when no limit order exists at the same side) or 1 (if at least one limit order exists at the same side)

$$Pr_{exec}(MO_{t_z}(A_i^{j*})) = 1 \Leftarrow lo_{t_{z-1}}(A_i^{j*} : A_i^j + TF^j = \min(A_i^j + TF^j)) > 0$$

Execution probability of *MOS*

$$Pr_{exec}(MO_{t_z}(B_i^{j*})) = 1 \Leftarrow lo_{t_{z-1}}(B_i^{j*} : B_i^j - TF^j = \max(B_i^j - TF^j)) > 0$$

4.2 Comparing payoffs

This section shows how traders in each period determine and compare their pay-offs from the available strategies in order to compute the submission probabilities. The execution probabilities of limit orders are uncertain since they depend on the submission probabilities of market orders in the subsequent periods. Since only market order or not trade strategies can be placed in t_3 , this uncertainty can be overcome by solving for the submission probabilities using backwards induction.

The payoff functions are defined as linear functions in terms of γ . Starting from $t_{T=3}$, payoff functions of *MO* and *NT* that result from the same order book state in t_{T-1} are equated with one another and solved for γ . The submission probability of a strategy $ST_{t_T}(P_i^j)$ is subsequently found by integrating $\gamma \sim U[0, 2]$ over the interval $[a, b] \subseteq [0, 2]$ that corresponds to where the payoff of $ST_{t_T}(P_i^j)$ is greater than or equal to the payoffs of other strategies in the comparison set. If two or more 'dominant' strategies in the comparison set have an equal payoff over $[a, b]$, then the submission probability will be equally distributed amongst the corresponding strategies over that interval. The payoff functions in t_T are described as follows:

$$\pi(ST_{t_z}(P_i^j)) = \begin{cases} \gamma_{t_z} - A_i^j - TF^j & | \ ST = MOB \\ B_i^j - \gamma_{t_z} - TF^j & | \ ST = MOS \\ 0 & | \ ST = NT \end{cases}$$

The submission probabilities in t_T can be computed by first describing a set $S = \{\pi(ST_{t_T}(P_i^j) \in \{MO_{t_T}(P_i^j), NT_{t_T}\})\}$ consisting of n payoff functions in t_T that all stem from the same order book state in t_{T-1} . Next, $g(\gamma) = \max_{\pi \in S}(\gamma)$ is defined as the function representing the maximum π -value achieved by any function in S at each point

$\gamma \in [0, 2]$ ¹¹. Finally, define the subset of S whose members equal $g(\gamma)$ over interval $I = [a, b] \subseteq [0, 2]$ as $S_I = \{\pi \in S | \pi(y) = g(\gamma)\}$. Denote the number of members within subset S_I as $m = |S_I|$, such that the submission probability for a strategy $ST_{t_T}(P_i^j)$ over $[a, b]$ can be defined as:

$$Pr_{subm}((ST_{t_T}(P_i^j))) = \frac{1}{m} \int_a^b d\gamma \quad \text{if } m \geq 1$$
¹²

In $t_Z < t_T$, the comparison set also includes the expected payoff for limit orders $\pi^e(LO_{t_z}(P_i^j))$. A limit order payoff is conditional on the execution probability of the limit order, which in turn depends on the submission probability of a corresponding market order in one of the subsequent periods $t_z \leq t_T$. The expected limit order buy and limit order sell payoffs can be formally defined as follows:

$$\begin{aligned} \pi^e(LO_{t_z}(B_i^j)) &= Pr_{exec}(LO_{t_z}(B_i^j))(\gamma_{t_z} - B_i^j + MF^j) \\ &= \left(\sum_{z=z+1}^{T-z} Pr_{subm}(MO_{t_z}(B_i^j))(\gamma_{t_z} - B_i^j + MF^j) \right) \end{aligned}$$

$$\begin{aligned} \pi^e(LO_{t_z}(A_i^j)) &= Pr_{exec}(LO_{t_z}(A_i^j))(A_i^j - \gamma_{t_z} + MF^j) \\ &= \left(\sum_{u=z+1}^{T-z} Pr_{subm}(MO_{t_u}(A_i^j))(A_i^j - \gamma_{t_z} + MF^j) \right) \end{aligned}$$

The expression above is not complete, as it does not take into account time priority. After all, if the limit order is placed at an existing queue in t_z , then the market orders placed at the same price and exchange in subsequent periods will not execute against the limit order placed in t_z as long as it is not first in the queue. We can show that a $MO_{t_z+c}(P_i^j)$ is executed against a $LO_{t_z}(P_i^j)$ in the previous period if the number of $MO_{t+u \in \{1, \dots, c\}}(P_i^j)$ is equal to $lo_{t_z}(P_i^j)$. In other words, a market order is executed against a corresponding limit order in the previous period t_z if the existing depth/queue at price P_i^j at the end of period t_z is fully eliminated when the market order is placed. Hence,

¹¹In financial economics, the function $g(\gamma)$ is also known as the *efficient frontier*. In mathematics, this is known as the *upper envelope*.

¹²This basically means that the submission probability is distributed equally over the m highest pay off strategies over interval $[a, b]$ if m is non zero. variable m must be non zero 1 since NT is always a possibility such that S_I is non empty

the expected limit order payoff can be defined as:

$$\pi^e(LO_{t_z}(P_i^j)) = \begin{cases} \left(\sum_{z=z+1}^{T-z} \Pr_{\text{subm}}(MO_{t_z}(B_i^j)) \middle| \sum_{u=1}^c MO_{t_z+u}(B_i^j) = lo_{t_z}(B_i^j) \right) (\gamma_{t_z} - B_i^j + MF^j) \\ \left(\sum_{z=z+1}^{T-z} \Pr_{\text{subm}}(MO_{t_z}(A_i^j)) \middle| \sum_{u=1}^c MO_{t_z+u}(A_i^j) = lo_{t_z}(A_i^j) \right) (A_i^j - \gamma_{t_z} + MF^j) \end{cases}$$

The submission probabilities in $t_z < t_T$ are subsequently computed by comparing the payoffs and integrating the gamma distribution over the cutoff gammas. Refer to appendix B for a worked out example.

Note that the submission probabilities described thus far are the probabilities relative to other strategies in the same comparison set, but not relative to all the possible outcomes in the concerning period. The latter is defined as the equilibrium submission probability of a strategy, denoted as $\Pr_{eqp}(ST_{t_z}(P_i^j))$, which is calculated as the product of the submission probabilities of all previous linked strategies. This metric is used as the building block for volume, order flow and market quality metrics. To formally define $\Pr_{eqp}(ST_{t_z}(P_i^j))$, let $k_l \in k = \{k_1, k_2, \dots, k_\omega\}$ be an element of a non ordered set k consisting of ω number of possible order book states in the tree. Define $L_{k_{(l-h) \geq 1}, k_l}$ as a vector consisting of the linked strategies between $ST^{k_{l-h}}$ and ST^{k_l} , and $N = |L_{k_{l-h}, k_l}|$ as the length of this vector such that:

$$\Pr_{eqp}(ST_{t_z}^{k_l}(P_i^j)) = \prod_{z=1}^N \Pr_{\text{subm}}(ST_{t_z}^{k_l}(P_i^j))$$

5 Market Indicators

5.1 Limit order flow

The limit order flow in an exchange can be interpreted as the number of limit orders that are submitted. The average limit order flow for exchange j , denoted as LO^j is computed as the sum of the weighted average limit order flow of each period, denoted as $LO_{t_z}^j$.

$$LO^j = \frac{1}{T} \sum_{z=1}^T LO_{t_z}^j$$

$LO_{t_z}^j$ is subsequently computed as the sum of all the equilibrium submission probabilities of limit orders at a particular period t_z . $LO_{t_z}^j$ is formally described by defining $S_{t_z}^{LO} = \{LO_{t_z}^{k_l}(P_i^j)\}$ as the set of all limit orders in period t_z and $N_{t_z}^{LO} = |S_{t_z}^{LO}|$ as the number of limit orders in this set such that

$$LO_{t_z}^j = \sum_{l=1}^{N_{t_z}^{LO}} \text{Pr}_{eqp} \left(LO_{t_z}^{k_l}(P_i^j) \right)$$

5.2 Trading volume

The trading volume is the number of submitted market orders. The average volume at exchange j , denoted as V^j , is computed as the sum the of the weighted average volume of each period denoted as $V_{t_z}^j$.

$$V^j = \frac{1}{T} \sum_{z=1}^T V_{t_z}^j$$

$V_{t_z}^j$ is subsequently computed as the sum of all the equilibrium submission probabilities of market orders at a particular period t_z . $V_{t_z}^j$ is formally described by defining $S_{t_z}^V = \{V_{t_z}^{k_l}(P_i^j)\}$ as the set of all market orders in period t_z and $N_{t_z}^V = |S_{t_z}^V|$ as the number of market orders in this set such that

$$V_{t_z}^j = \sum_{l=1}^{N_{t_z}^V} \text{Pr}_{eqp} \left(V_{t_z}^{k_l}(P_i^j) \right)$$

The market share of each exchange j is denoted as MS^j and represents the relative share of the total trading volume across all exchanges. Let N^{exch} be the number of exchanges such that

$$MS^j = \frac{V^j}{\sum_{j=1}^{N^{exch}} V^j}$$

5.3 Spread

The raw spread is the difference between the best available ask and bid price. When the books are empty, the spread is assumed to be 0.05 (5 ticks). The average spread $Spread^j$ is computed as the weighted average spread of each period.

$$Spread^j = \frac{1}{T} \sum_{z=1}^T Spread_{t_z}^j$$

The spread per period t_z is computed by summing the expected spread of each order book state in equilibrium. Denote $N_{t_z}^{OB} = N_{t_z}^{LO} + N_{t_z}^V$ as the number of possible order book states in each period t_z , and A_b^{j,k_l} , B_b^{j,k_l} as the best ask price and best bid price respectively corresponding to order book state k of exchange j .

$$Spread_{t_z}^j = \sum_{l=1}^{N_{t_z}^{OB}} \Pr_{eqp} \left(OB_{t_z}^{j,k_l} \right) \left(A_b^{j,k_l} - B_b^{j,k_l} \right)$$

5.4 Depth at best bid ask

The depth at the best bid ask price shows the number of limit orders at the best bid and best ask price. The average depth $BBADepth^j$ is computed in a similar way as the spread mentioned in the previous subsection.

$$BBADepth^j = \frac{1}{T} \sum_{z=1}^T BBADepth_{t_z}^j$$

$$BBADepth_{t_z}^j = \sum_{l=1}^{N_{t_z}^{OB}} \Pr_{eqp} \left(OB_{t_z}^{j,k_l} \right) \left(lo(A_b^{j,k_l}) + lo(B_b^{j,k_l}) \right)$$

6 Results

This section describes the results with respect to liquidity, volume, depth, spread and net revenue of both exchanges. First, two identical exchanges without fees are compared to gain a baseline understanding. Next, the comparison is extended to a setting where one operates on the make-take-fees, and the other on take-make-fees. The trading fees are based on the latest equity fee regimes from major US exchanges as a reference. The standard (non-tiered) fees are used from NYSE, NASDAQ and IEX. Each of these fee structures are compared to the inverted fee structures of EDGA, BATS Y and NASDAQ BX. The results of this section can be found in the tables of Appendix A.

Proposition 1. *Two competing exchanges that operate without fees and maintain identical order books will have:*

- *identical average volume and limit order flow*
- *identical average spread and depth*
- *increased average volume and limit order flow at the inner prices of the order book*

The first two statements are straightforward to understand since traders are indifferent between identical exchanges. The third statement is explained by the concept of undercutting (or front running) in a three-plus period trading game: an inner priced *LO* placed in t_2 can undercut an outer priced *LO* placed in t_1 . An inner priced *LO* placed in t_1 can never be undercut in the absence of trading fees due to time price priority. Consequently leading to an increased execution probability of the inner priced *LO* compared to the outer priced *LO*. In other words, for outer priced orders the decrease in execution probability does not offset the price improvement to be gained in the next period. However, the offset is valid for all subsequent periods that follow due to undercutting. Meaning that trading activity will always be concentrated at the inner quotes in a three plus period model when no frictions are present.

Proposition 2. *In a competitive setting where two exchanges have identical order books, one operating on make-take-fees and the other on take-make-fees, then the latter will experience:*

- *less average limit order flow and volume at the outer price levels*
- *more average limit order flow and volume at the inner price levels*
- *more average limit order flow and volume overall*
- *narrower average bid-ask spread and enhanced depth at the best bid and ask*

In all scenarios, the take-make venue captures nearly all order flow and volume from the make-take venue at the inner quotes, but not at the outer quotes. The same reasoning can be followed outlined in Proposition 1. Traders place orders at the inner quotes of the inverted venue because the narrower cum fee spread allows them to gain queue priority. In contrast, make-take venues have more spaced-out cum fee price grids which allows them to capture a larger proportion of traders at the outer quotes. Based on the explanation in subsection 4.1.1., the arriving trader will logically place a limit order in t_{m-1} at the outer quote of the make-take venue if no limit orders at the same side exists across all venues. Since this model only assumes one arriving trader per period in a three period game, there will always be some possible market state in t_2 where an outer quoted limit order at the make-take venue has a non zero execution probability. It is important to stress out that this observation is heavily dependent on these simplified assumptions and that generalizations should be made with caution.

Proposition 3. *In a competitive setting where two exchanges have identical order books, one operating on make-take-fees and the other on take-make-fees, then the sensitivity of fees on trading activity:*

- *is inelastic at the take- make venue*
- *shows more in-elasticity with respect to take-fees within an exchange*
- *shows more in-elasticity with respect to take-fees across exchanges*

Regardless of which make-fees are set on either exchange - and assuming a capped fee of 0.003, if the take-fee of one exchange is lower than the take-fee of another, then the exchange with the lower take-fee will always attract more order flow by setting its take-fee slightly below the take-fee of the competition exchange. To illustrate this,

suppose Exchange I sets its fees to $MF^I, TF^I = \{0.002, 0.003\}$. Conditional on these fees, the competition exchange can optimize net revenue by setting their fees to $MF^{II}, TF^{II} = \{-0.003, 0.0029\}$. This means that the competition exchange could charge a fee to both liquidity takers and providers, without losing much on trading activity.¹³ This does not contradict propositions 1 and 2, as the price grid for both liquidity providers and liquidity takers becomes narrower resulting in more overall trading activity compared to the make-take venue. However, charging a fee to both liquidity makers and takers does not seem to align with reality. The implications are interesting. Especially when exploring the relationship between trading fees and volume. And the impact this has on the net revenue between the different fee structures. Based on the propositions mentioned so far, inverted venues are hypothesized to always be the better choice in an unrestricted fee regime¹⁴. Hence, exchanges will compete by setting lower take-fees, regardless of which make-fees they or their competitor set.

¹³This example can be replicated. In equilibrium, the make- take venue has an average limit order flow and volume of 0.0292 and 0.0143 respectively, whereas the fee optimizing exchange has an average limit order flow and volume of 0.4898 and 0.3616 respectively

¹⁴In the presence of unrestricted fees, the competition exchange in the example above can set their make-fee to an astonishing -0.17 and reach a net revenue of 0.0342. Decreasing the make-fee further would lead to a negative impact on net revenue

7 Conclusion

This is a model that frames the evolution of order books of two competing exchanges. Inverted venues have narrower price grids which allow traders to trade at sub-tick prices and front run orders placed at make-take venues. Therefore, most liquidity provision and volume happens at the inner quotes of inverted venues. This study also shows that make-take venues experience more order flow and volume at the outer quotes as every opening book in t_2 will have an empty buy or side, presenting the opportunity to provide liquidity at the outer quotes of the make-take venue without losing out too much on execution probability.

Despite a significant larger market share captured by inverted venues as predicted by the model, empirical data shows that the majority of trading activity remains at the make-take venues. The discrepancy between real world outcomes and the model suggests that the assumptions made in this model are either not sufficiently realistic, incorrect or a combination of both. Further research can be conducted by performing a sensitivity analysis on the asset price, assuming skewed or Gaussian gamma distributions, adding more traders and assuming more complex fee structures, different order types or including additional market participants such as brokers and governmental bodies (i.e. accounting for transaction tax)¹⁵.

¹⁵refer to section Reflection

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Appendix A

Table 1: Trading activity and market quality results in a competitive setting of two identical exchanges in a three-period model without fees.

	Exchange I	Exchange II		
Take fee	0,00	0,00	difference	
Make fee	0,00	0,00	Δ	%
	inner prices			
Liquidity	0,3657	0,3657	0,00	0,00
Volume	0,1816	0,1816	0,00	0,00
	outer prices			
Liquidity	0,0124	0,0124	0,00	0,00
Volume	0,0061	0,0061	0,00	0,00
	total			
Liquidity	0,3781	0,3781	0,00	0,00
Volume	0,1877	0,1877	0,00	0,00
Spread	0,0401	0,0401	0,00	0,00
BBAdepth	0,5042	0,5042	0,00	0,00

Table 2: Trading activity and market quality results in a competitive setting of two exchanges in a three-period model. NYSE operates on maker taker fees and is compared separately with NASDAQ BX, EDGA and BATS Y which operate on inverted maker taker fees. We compute the market metrics for each price level and show the absolute and relative difference of each metric between each inverted venue and NYSE

	NYSE	NASDAQ BX			NYSE	EDGA			NYSE	BATS Y		
Take fee	0.003	-0.0018	difference		0.003	-0.0014	difference		0.003	-0.0002	difference	
make fee	0.00275	-0.003	Δ	%	0.00275	-0.003	Δ	%	0.00275	-0.002	Δ	%
	Inner prices				Inner prices				Inner prices			
Liquidity	0.0014	0.4860	-0.4846	-99.71%	0.0018	0.4860	-0.4842	-99.63%	0.0016	0.4874	-0.4858	-99.68%
Volume	0.0007	0.3613	-0.3606	-99.81%	0.0009	0.3611	-0.3602	-99.75%	0.0008	0.3617	-0.3609	-99.79%
Net revenue	1.75E-07	4.34E-04	-4.33E-04	-99.96%	2.25E-07	5.78E-04	-5.78E-04	-99.96%	1.93E-07	6.51E-04	-6.50E-04	-99.97%
	outer prices				outer prices				outer prices			
Liquidity	0.0244	0.0056	0.0188	337.86%	0.0246	0.0052	0.0194	375.64%	0.0237	0.0047	0.0190	402.19%
Volume	0.0119	0.0027	0.0092	333.86%	0.0120	0.0025	0.0095	371.51%	0.0116	0.0023	0.0093	398.51%
Net revenue	2.98E-06	3.30E-06	-3.17E-07	-9.61%	3.01E-06	4.08E-06	-1.07E-06	-26.33%	2.90E-06	4.19E-06	-1.29E-06	-30.76%
	total				total				total			
Liquidity	0.0258	0.4917	-0.4659	-94.75%	0.0264	0.4912	-0.4648	-94.62%	0.0253	0.4921	-0.4668	-94.85%
Volume	0.0126	0.3640	-0.3514	-96.54%	0.0129	0.3637	-0.3508	-96.45%	0.0124	0.3640	-0.3516	-96.59%
Net revenue	3.15E-06	4.37E-04	-4.34E-04	-99.28%	3.15E-06	5.82E-04	-5.79E-04	-99.46%	3.15E-06	6.55E-04	-6.52E-04	-99.52%
Spread	0.0497	0.0353	0.0144	40.79%	0.0497	0.0353	0.0144	40.79%	0.0497	0.0352	0.0145	29.18%
BBAdepth	0.0317	0.7403	0.7085	-95.71%	0.032	0.7400	0.7080	-95.68%	0.0304	0.7423	0.7119	-95.9%

Table 3: Trading activity and market quality results in a competitive setting of two exchanges in a three-period model. NASDAQ operates on maker taker fees and is compared separately with NASDAQ BX, EDGA and BATS Y which operate on inverted maker taker fees. We compute the market metrics for each price level and show the absolute and relative difference of each metric between each inverted venue and NASDAQ

	NASDAQ	NASDAQ BX			NASDAQ	EDGA			NASDAQ	BATS Y		
Take fee	0.003	-0.0018	difference		0.003	-0.0014	difference		0.003	-0.0002	difference	
make fee	0.0018	-0.003	Δ	%	0.0018	-0.003	Δ	%	0.0018	-0.002	Δ	%
	Inner prices				Inner prices				Inner prices			
Liquidity	0.0001	0.4869	-0.4868	-99.99%	0.0006	0.4869	-0.4863	-99.88%	0.0006	0.4883	-0.4877	-99.87%
Volume	3.25E-05	0.3621	-0.3621	-99.99%	0.0003	0.3619	-0.3616	-99.92%	0.0003	0.3625	-0.3622	-99.91%
Net revenue	3.90E-08	4.34E-04	-4.34E-04	-99.99%	3.54E-07	5.79E-04	-5.79E-04	-99.94%	3.77E-07	6.53E-04	-6.53E-04	-99.94%
	outer prices				outer prices				outer prices			
Liquidity	0.0230	0.0064	0.0166	259.65%	0.0232	0.0059	0.0173	290.77%	0.0224	0.0052	0.0172	330.17%
Volume	0.0112	0.0032	0.0080	256.43%	0.0114	0.0029	0.0085	287.44%	0.0109	0.0026	0.0083	327.10%
Net revenue	1.35E-05	3.79E-06	9.71E-06	256.43%	1.36E-05	4.69E-06	8.91E-06	190.58%	1.31E-05	4.61E-06	8.49E-06	184.74%
	total				total				total			
Liquidity	0.0231	0.4934	-0.4703	-95.33%	0.0238	0.4928	-0.4691	-95.17%	0.0230	0.4935	-0.4705	-95.34%
Volume	0.0113	0.3651	-0.3538	-96.91%	0.0116	0.3648	-0.3532	-96.82%	0.0112	0.3651	-0.3539	-96.93%
Net revenue	1.36E-05	4.38E-04	-4.25E-04	-96.91%	1.39E-05	5.84E-04	-5.70E-04	-97.62%	1.34E-05	6.57E-04	-6.44E-04	-97.95%
Spread	0.0497	0.0352	-0.0145	41.19%	0.0497	0.0352	-0.0145	41.19%	0.0497	0.0352	-0.0145	41.19%
BBAdepth	0.0298	0.7422	-0.7124	-95.99%	0.0301	0.7420	-0.7119	-95.99%	0.0285	0.7439	-0.7154	-96.17%

Table 4: Trading activity and market quality results in a competitive setting of two exchanges in a three-period model. IEX operates on maker taker fees and is compared separately with NASDAQ BX, EDGA and BATS Y which operate on inverted maker taker fees. We compute the market metrics for each price level and show the absolute and relative difference of each metric between each inverted venue and IEX

	IEX	NASDAQ BX			IEX	EDGA			IEX	BATS Y		
Take fee	0.001	-0.0018	difference		0.001	-0.0014	difference		0.001	-0.0002	difference	
make fee	0.0004	-0.003	Δ	%	0.0004	-0.003	Δ	%	0.0004	-0.002	Δ	%
	Inner prices				Inner prices				Inner prices			
Liquidity	0.0006	0.4882	-0.4876	-99.88%	0.0009	0.4882	-0.4873	-99.82%	0.0008	0.4896	-0.4888	-99.84%
Volume	0.0003	0.3631	-0.3628	-99.92%	0.0004	0.3630	-0.3625	-99.88%	0.0004	0.3636	-0.3632	-99.89%
Net revenue	1.78E-07	4.36E-04	-4.36E-04	-99.96%	2.63E-07	5.81E-04	-5.81E-04	-99.95%	2.40E-07	6.54E-04	-6.54E-04	-99.96%
	outer prices				outer prices				outer prices			
Liquidity	0.0210	0.0053	0.0157	296.61%	0.0213	0.0050	0.0162	322.59%	0.0204	0.0044	0.0160	363.91%
Volume	0.0103	0.0026	0.0077	293.96%	0.0104	0.0025	0.0079	319.95%	0.0100	0.0022	0.0078	361.66%
Net revenue	6.19E-06	3.14E-06	3.05E-06	96.98%	6.25E-06	3.97E-06	2.28E-06	57.48%	6.00E-06	3.90E-06	2.10E-06	53.89%
	total				total				total			
Liquidity	0.0216	0.4935	-0.4719	-95.62%	0.0221	0.4932	-0.4711	-95.52%	0.0212	0.494	-0.4728	-95.71%
Volume	0.0106	0.3658	-0.3552	-97.10%	0.0108	0.3655	-0.3547	-97.05%	0.0104	0.3658	-0.3554	-97.16%
Net revenue	6.36E-06	4.39E-04	-4.39E-04	-98.55%	6.48E-06	5.85E-04	-5.85E-04	-98.89%	6.24E-06	6.58E-04	-6.58E-04	-99.05%
Spread	0.0497	0.0352	-0.0145	41.19%	0.0497	0.0352	-0.0145	41.19%	0.0497	0.0351	-0.0146	41.60%
BBAdepth	0.0272	0.7427	-0.7155	-96.34%	0.0275	0.7426	-0.7151	-96.30%	0.0258	0.7447	-0.7189	-96.54%

Table 5: This table reports the equilibrium outcome of Exchange I and Exchange II in a three period model. We compute the results for a changing MF and TF of exchange I with constant MF=TF=0 of Exchange II. The results account for order flow, volume, market quality and net revenue and are subsequently compared to when both exchanges have no fees (as per column 2). The absolute and relative difference between each setting and the benchmark are computed, and is shown under 'difference'

no fees		make-take fees						inverted make-take fees					
Market I TF	0,000	0,001	difference		0,002	difference		-0.001	difference		-0.001	difference	
Market I MF	0,000	0,001	Δ	%	0,001	Δ	%	-0.001	Δ	%	-0.002	Δ	%
Volume ^I	0,1877	0,0091	-0,1786	-95,2%	0,009	-0,1787	-95,2%	0,3674	0,1797	95,7%	0,3678	0,1801	96,0%
Volume ^{II}	0,1877	0,3672	0,1795	95,6%	0,3672	0,1795	95,6%	0,0088	-1789	-95,3%	0,0085	-0,1792	-95,5%
Total volume	0,3754	0,3763	0,0009	0,2%	0,3762	0,0008	0,2%	0,3762	0,0008	0,2%	0,3763	0,0009	0,2%
Market share ^I	0,5	0,0241	-0,4759	-95,2%	0,0239	-0,4761	-95,2%	0,9765	0,4765	95,3%	0,9773	0,4773	95,5%
Market share ^{II}	0,5	0,9759	0,4759	95,2%	0,9761	0,4761	95,2%	0,0235	-0,4765	-95,3%	0,0063	-0,4937	-98,7%
Liquidity ^I	0,3781	0,0185	-0,3596	-95,1%	0,0183	-0,3598	-95,2%	0,4957	0,1176	31,1%	0,4957	0,1176	31,1%
Liquidity ^{II}	0,3781	0,4959	0,1178	31,2%	0,496	0,1179	31,2%	0,018	-0,3601	-95,2%	0,0174	-0,3607	-95,4%
Total liquidity	0,7562	0,5144	-0,2418	-32,0%	0,5143	-0,2419	-32,0%	0,5137	-0,2425	-32,1%	0,5131	-0,2431	-32,1%
Spread ^I	0,0401	0,0488	0,0087	21,7%	0,0498	0,0097	24,2%	0,0351	-0,005	-12,5%	0,0351	-0,005	-12,5%
Spread ^{II}	0,0401	0,0342	-0,0059	-14,7%	0,0351	-0,005	-12,5%	0,0498	0,0097	24,2%	0,0498	0,0097	24,2%
Depth ^I	0,5042	0,004	-0,5002	-99,2%	0,0226	-0,4816	-95,5%	0,7469	0,2427	48,1%	0,7465	0,2423	48,1%
Depth ^{II}	0,5042	0,738	0,2338	46,4%	0,7468	0,2426	48,1%	0,0224	-0,4818	-95,6%	0,0219	-0,4823	-95,7%
Net revenue ^I	0	0	0	-	1,80E-05	1,80E-05	-	0	0	-	7,36E-04	7,36E-04	-
Net revenue ^{II}	0	0	0	-	0	0	-	0	0	-	0	0	-

Appendix B

This section solves the equilibrium submission probabilities for exchange I, with exchange I having no fees and exchange II having make-take-fees $\{MF^{II}, TF^{II}\} = \{0.001, 0.001\}$. At the beginning of t_1 an arriving trader wants to know what the possible outcomes of the order books are based on the opening market. Since the books open empty, the only strategies he can choose from are limit orders or not trade. Consequently, the possible market states at the end of t_1 are:

$$[0, 0, 0, 0 - 0, 0, 0, 0] \left\{ \begin{array}{ll} [0, 0, 0, 0 - 0, 0, 0, 0] & NT \\ [1, 0, 0, 0 - 0, 0, 0, 0] & MO_{t_1}(A_2^I) \\ [0, 1, 0, 0 - 0, 0, 0, 0] & LO_{t_1}(A_1^I) \\ [0, 0, 1, 0 - 0, 0, 0, 0] & LO_{t_1}(B_1^I) \\ [0, 0, 0, 1 - 0, 0, 0, 0] & LO_{t_1}(B_2^I) \\ [0, 0, 0, 0 - 1, 0, 0, 0] & LO_{t_1}(A_2^{II}) \\ [0, 0, 0, 0 - 0, 1, 0, 0] & LO_{t_1}(A_1^{II}) \\ [0, 0, 0, 0 - 0, 0, 1, 0] & LO_{t_1}(B_1^{II}) \\ [0, 0, 0, 0 - 0, 0, 0, 1] & LO_{t_1}(B_2^{II}) \end{array} \right.$$

Figure 1: Possible order book states at the end of t_1

From here on, we will go further with the order book state $[0, 0, 0, 0 - 0, 1, 0, 0]$. At t_2 , the books open $[0, 0, 0, 0 - 0, 1, 0, 0]$. An arriving trader can choose to not trade, submit a market order buy at the inner quote of exchange II, or submit limit orders that have a non-zero execution probability based on the current order book: placing a limit order sell at either exchange is not feasible, since there already exists a limit order sell at the lowest cum fee bid price; placing a limit order buy at each price level at either exchanges is also possibility, since there is no limit orders buy yet implying time priority of the buy order. Hence, the possible order book states at the end of t_2 are:

$$[0, 0, 0, 0 - 0, 0, 0, 0] \left\{ \begin{array}{ll} [0, 0, 0, 0 - 0, 1, 0, 0] & NT_{t_2}(0) \\ [0, 0, 0, 0 - 0, 0, 0, 0] & MO_{t_2}(A_1^{II}) \\ [0, 0, 1, 0 - 0, 1, 0, 0] & LO_{t_2}(B_1^I) \\ [0, 0, 0, 1 - 0, 1, 0, 0] & LO_{t_2}(B_2^I) \\ [0, 0, 0, 0 - 0, 1, 1, 0] & LO_{t_2}(B_1^{II}) \\ [0, 0, 0, 0 - 0, 1, 0, 1] & LO_{t_2}(B_2^{II}) \end{array} \right.$$

Figure 2: Possible order book states at the end of t_2

In a similar way, the possible market states in t_3 are mapped as follows:

$$\begin{aligned} [0, 0, 0, 0 - 0, 1, 0, 0] & \left\{ \begin{array}{ll} [0, 0, 0, 0 - 0, 1, 0, 0] & NT_{t_3}(0) \\ [0, 0, 0, 0 - 0, 0, 0, 0] & MO_{t_3}(S_1^{II}) \end{array} \right. \\ [0, 0, 0, 0 - 0, 0, 0, 0] & \left\{ [0, 0, 0, 0 - 0, 0, 0, 0] \quad NT_{t_3}(0) \right. \\ [0, 0, 1, 0 - 0, 1, 0, 0] & \left\{ \begin{array}{ll} [0, 0, 1, 0 - 0, 1, 0, 0] & NT_{t_3}(0) \\ [0, 0, 1, 0 - 0, 0, 0, 0] & MO_{t_3}(S_1^{II}) \\ [0, 0, 0, 0 - 0, 1, 0, 0] & MO_{t_3}(B_1^I) \end{array} \right. \\ [0, 0, 0, 1 - 0, 1, 0, 0] & \left\{ \begin{array}{ll} [0, 0, 0, 1 - 0, 1, 0, 0] & NT_{t_3}(0) \\ [0, 0, 0, 1 - 0, 0, 0, 0] & MO_{t_3}(S_2^{II}) \\ [0, 0, 0, 0 - 0, 1, 0, 0] & MO_{t_3}(B_2^I) \end{array} \right. \\ [0, 0, 0, 0 - 0, 1, 1, 0] & \left\{ \begin{array}{ll} [0, 0, 0, 0 - 0, 1, 1, 0] & NT_{t_3}(0) \\ [0, 0, 0, 0 - 0, 0, 1, 0] & MO_{t_3}(S_1^{II}) \\ [0, 0, 0, 0 - 0, 1, 0, 0] & MO_{t_3}(B_1^I) \end{array} \right. \\ [0, 0, 0, 0 - 0, 1, 0, 1] & \left\{ \begin{array}{ll} [0, 0, 0, 0 - 0, 1, 0, 1] & NT_{t_3}(0) \\ [0, 0, 0, 0 - 0, 0, 0, 1] & MO_{t_3}(S_1^{II}) \\ [0, 0, 0, 0 - 0, 1, 0, 0] & MO_{t_3}(B_2^{II}) \end{array} \right. \end{aligned}$$

Figure 3: Possible order book states at the end of t_3

After the trader has mapped all possible order book states for the entire trading game, he can now start comparing the payoffs for each of these strategies and working his way backwards to compute the real submission and execution probabilities. .

At t_3 , the strategies are grouped according to the same order book state they descended from in t_2 . First, the payoffs of the strategies are defined in function of γ . Next, the payoffs are compared with one another within each group and the corresponding submission probabilities per strategy are derived. This is done by integrating the gamma distribution over the interval where the payoff of the strategy is maximized across all the strategies in the comparison set. The integration is subsequently divided by the number of equal payoffs in the same interval. The table below shows the submission probability for each strategy in t_3 .

opening book in t_3	strategy $ST_{t_3,i}^j$	pay off $\pi_{t_3}^e$	$\gamma_{t_3} \pi_{t_3}^e = \max$	$Pr(ST_{t_3,i}^j)$ ¹⁶
[0, 0, 0, 0 – 0, 1, 0, 0]	$MOB(A_1^{II})$	$\gamma_{t_3} - 1.005$	[1.005, 2]	.4975
	MOS	/	/	0
	NT	0	[0, 1.005]	0.5025
[0, 0, 0, 0 – 0, 0, 0, 0]	MOB	/	/	0
	MOS	/	/	0
	NT	0	[0, 2]	1
[0, 0, 1, 0 – 0, 1, 0, 0]	$MOB(A_1^{II})$	$\gamma_{t_3} - 1.005$	[1.005, 2]	0.4975
	$MOS(B_1^I)$	$0.994 - \gamma_{t_3}$	[0, 0.994]	0.497
	NT	0	[.994, 1.005]	.0055
[0, 0, 0, 1 – 0, 1, 0, 0]	$MOB(A_1^{II})$	$\gamma_{t_3} - 1.005$	[1.005, 2]	0.4975
	$MOS(B_2^I)$	$0.984 - \gamma_{t_3}$	[0, 0.984]	0.4920
	NT	0	[.984, .994]	.0105
[0, 0, 0, 0 – 0, 1, 1, 0]	$MOB(A_1^{II})$	$\gamma_{t_3} - 1.005$	[1.005, 2]	0.4975
	$MOS(B_1^{II})$	$0.995 - \gamma_{t_3}$	[0, 0.995]	0.4975
	NT	0	[0.995, 1.005]	0.005
[0, 0, 0, 0 – 0, 1, 0, 1]	$MOB(A_1^{II})$	$\gamma_{t_3} - 1.005$	[1.005, 2]	.4975
	$MOS(B_2^{II})$	$0.985 - \gamma_{t_3}$	[0, 0.985]	.4925
	NT	0	[.985, 1.005]	.015

Table 6: submission probabilities in t_3

At t_2 , the strategies to be compared also include limit orders. The payoff of a limit order is conditional on it being executed against a market order in a subsequent period. In

other words, the limit order payoff must take into account the submission probability of a corresponding market order that we found above. The (expected) payoffs are subsequently compared and submission probabilities are derived in the same way as in t_3 . The table below shows the submission probabilities in t_3 .

opening book in t_2	strategy $ST_{t_2,i}^j$	pay off $\pi_{t_2}^e$	$\gamma_{t_2} \pi_{t_3}^e = max$	$Pr(ST_{t_2,i}^j)$ ¹⁷
[0, 0, 0, 0 – 0, 1, 0, 0]	$MOB(A_1^{II})$	$\gamma_{t_2} - 1.005$	[1.0253, 2]	.48735
	$LOB(B_2^I)$	$.4920(\gamma_{t_2} - .984)$	[.984, 1.0253]	.02065
	$LOB(B_2^{II})$	$.4925(\gamma_{t_2} - .985)$	/	0
	$LOB(B_1^I)$	$.497(\gamma_{t_2} - .994)$	/	0
	$LOB(B_1^{II})$	$.4975(\gamma_{t_2} - .985)$	/	0
	NT	0	[0, .984]	.492

Table 7: submission probabilities in t_2

The same reasoning applies to deriving the submission probabilities of limit orders in t_1 . As mentioned in the previous paragraph, this is done by comparing the payoffs, which in turn is based on the submission probability of a corresponding market order in the subsequent period. The difference now is that there could be multiple possible market orders executed against the limit order. After all, a limit order placed in t_1 has some chance to be executed against a market order in t_2 or t_3 . The execution probability for $LOS_{t_1,1}^{II}$ specifically, is the probability of a MOB_1^{II} being placed in t_2 plus the probability of a MOB_1^{II} not being placed in t_2 but in t_3 .

$$\begin{aligned}
\Pr_{\text{subm}}(LOS_{t_1}(A_1^{II})) &= \Pr_{\text{subm}}(MOB_{t_2}(A_1^{II})) + (1 - \Pr_{\text{subm}}(MOB_{t_2}(A_1^{II}))) \cdot \Pr_{\text{subm}}(MOB_{t_3}(A_1^{II})) \\
&= 0.48735 + (1 - 0.48735) \cdot 0.4975 \\
&= 0.7424
\end{aligned}$$

The expected pay off for $LOS_{t_1,1}^{II}$ is therefore:

$$\begin{aligned}
\pi^e(LOS_{t_1}(A_1^{II})) &= (1.005 - \gamma_{t_1}) \cdot 0.7424 \\
&= 0.7461 - 0.7424\gamma_{t_1}
\end{aligned}$$

The trader applies the same computations, as described above, for all the other possible limit orders in t_1 and comes to the following results.

opening book in t_1	strategy $ST_{t_1,i}^j$	payoff $\pi_{t_1}^e$	$\gamma_{t_1} \pi_{t_1}^e = \max$	$Pr(ST_{t_1,i}^j)$
[0, 0, 0, 0 – 0, 0, 0, 0]	NT	0	/	0
	$(LOS(A_{1,2}^I))$	$0.4960 - 0.4882\gamma$	$[0.9839, 1]$	0.00806
	$(LOS(A_{1,1}^I))$	$0.4975 - 0.4945\gamma$	/	0
	$(LOB(B_{1,1}^I))$	$0.4945\gamma - 0.4915$	/	0
	$(LOB(B_{1,2}^I))$	$0.4882\gamma - 0.4804$	$[1, 1.0161]$	0.00806
	$(LOS(A_{1,2}^{II}))$	$0.4971 - 0.4898\gamma$	/	0
	$(LOS(A_{1,1}^{II}))$	$0.7461 - 0.7424\gamma$	/	0.49194
	$(LOB(B_{1,1}^{II}))$	$0.7424\gamma - 0.7387$	$[1.0161, 2]$	0.49194
	$(LOB(B_{1,2}^{II}))$	$0.4898\gamma - 0.4824$	/	0

Table 8: this table shows the equilibrium submission strategies with their corresponding payoffs, gamma interval and submission probabilities

Reflection

Limitations of the research

Certain assumptions did not reflect reality. For example, the model assumes one arriving trader per period. Whereas in reality many arriving traders can arrive each period or none at all. This dynamic process can be approximated by incorporating some stochastic process. Another limitation is the uniform gamma-distribution along the interval of $[0,2]$. This interval could be bigger, and the distribution can be skewed or Gaussian for example.

Furthermore, we assumed that the trader direct market access to the exchanges. In reality a significant portion of traders place their orders through intermediaries like brokers, who then route these orders to exchanges on their behalf. This leads to the principal-agent problem and could have a significant impact on the results. Other market participants such as HFT's, hedge funds, large block traders, market makers, clearinghouses, regulatory frictions etc. are also not taken into account.

Possibilities for further research

Financial Transaction Tax

Introducing financial transaction taxes (FTTs) is an easy and straightforward way to improve the realism of the model. The FTT expense is calculated as a percentage of the transaction price. We can denote FTT as $t \cdot P$, where t stands for the tax rate and P for the transaction price. A sensitivity analysis could be performed based on the tax rates per jurisdiction. For Belgian residents for example, FTTs for non-REIT stocks amounts to 0.35% of the transaction price. So, $t \cdot p = 0.0035 \cdot p$.

$$\begin{aligned}
\pi(MOB) &= \gamma - P \cdot (1 + t) - TF \\
\pi(MOS) &= P \cdot (1 - t) - TF - \gamma \\
\pi(LOB) &= \Pr(MOS) \cdot (\gamma - P \cdot (1 + t) + MF) \\
\pi(LOS) &= \Pr(MOB) \cdot (P \cdot (1 - t) + MF - \gamma) \\
\pi(NT) &= 0
\end{aligned}$$

By introducing FTT's, pay off functions of market orders will shift downwards, resulting in lower gamma intervals and thus a decrease in equilibrium submission probabilities relative to NT. Due to this decrease, the execution probabilities of limit orders in the previous period will also be lower, resulting in a slope that is 'less steep'. However, this also means that the downwards shift of the pay off function will be relatively smaller compared with the market order payoffs. Due to this ambiguous effect, the movement of equilibrium limit order submission probabilities is pertinent relative to market orders.

Dividing the trading population into subgroups

We follow the same assumption as [Panayides et al. \(2023\)](#) that market participants are simultaneously buyers or sellers. By also introducing participants who are either buyers or sellers, as assumed by [Parlour \(1998\)](#), the trading population could be subsequently divided into 3 subgroups:

- (a) Participants are simultaneously buyer and seller
- (b) Participants are buyers only
- (c) Participants are sellers only

In case of (a), participants are traders who place orders purely with the intention to make money. There is no utility to be gained by having the asset. The sole purpose is to trade it at a later date in the hopes of making money from the process. Think of market makers, speculators, HFT's, hedge funds, arbitrageurs etc.

In case of (b) or (c), positions are taken because taking a position in the underlying asset has real utility for the participant. This can be an acquirer who takes a toehold in a hostile takeover, a seller in a leveraged or management buy-out, a coffee producer buying futures to hedge against price fluctuations etc.

The main advantage of including this groups is that in these instances, the fundamental asset price is known and the trading period is finite. The distribution of these trading populations into subgroup a, b or c could be $1/4$, $1/4$ and $1/2$ for example to simulate a market selloff.

By including participants who can only sell or buy, these participants are conjectured to more likely refrain from trading compared to traders who can both buy and sell. This leads to a decrease in trading volume and limit order flow relative to when there are only traders of type (a) present.

Given the substantives of this assumption, it is worthwhile to delve deeper into this conjecture in future research.